

Cross-Ownership in Inter-Corporate Loans in China

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Abstract: This paper investigates how firms mitigate information asymmetry in inter-corporate loans via cross-ownership linkages between borrowers and lenders. We find that cross-owned firms borrow at significantly lower interest rates and are less likely to be required to pledge collaterals. Economically, cross-ownership reduces loan rates by 3.4 percent and lowers the collateral requirements by almost 50 percent. We further find that the beneficial effect of cross-ownership on inter-corporate loans is greater for firms with poorer credit access, in provinces with less developed financial markets or weaker legal protections, or during times of tight monetary policy.

Keywords: Inter-corporate loans, informal financing, cross-ownership, China

JEL: G20, G30, H81, L10

1. Introduction

Ownership structure is an important factor in influencing the cost and debt structure of borrowings via bank loans (Lin, Ma, Malatesta and Xuan, 2011, 2012, 2013). The effect of ownership structure on non-bank debt borrowing, however, is not examined in the extant literature. We are the first, to the best of our knowledge, to examine this important issue since informal financing is prevalent in the emerging markets and is a contributing factor for financial stability. Studies show that informal financial market emerges when firms experience greater difficulties in accessing to formal financing (Levine, 1997; Demirguc-Kunt and Maksimovic, 1998; Allen, Qian and Qian, 2005). Many firms, especially the private and smaller firms, rely more on informal credit to finance their investment and operation. This phenomenon is particularly salient in an emerging economy like China where the development of the formal financial market lags behind the economic development. As a result, the formal banking is increasingly focusing on the financial needs of the large, state-owned or public enterprises, which leaves many smaller, non-state-owned and private borrowers unserved. Rating agency S&P estimated that the outstanding volume of shadow banking activities amounted to US\$ 3.7 trillion by the end of 2012, which equals to 44 percent of the gross domestic product in China.

In informal borrowings, firms have to pay substantially higher cost to obtain the credit than from formal financing, i.e. due to more severe information asymmetry issue. As a result, these informal financing activities add to the instability of the financial system, and consequently the economic development. The regulators are also concerned that borrowing from the informal sector imposes severe financial burdens for borrowing corporations, which may lead to more risk-taking behaviors that harm the healthiness of the economy. Among the informal financing activities, this paper focuses on inter-corporate loans, which, according to the People's Bank of China (PBOC), amount to US\$ 0.25 trillion by 2015 and become the second largest source of debt financing for corporations after commercial banks.

With inter-corporate loans, a company provides the funds but, to circumvent a ban on direct lending to other firms, it designates a commercial bank to lend the money to a specific borrower. The lending firm stipulates the amount, tenor, and rate of the loan, while the banks only earn fees from both sides without the loans showing up on their

balance sheets. Without the banks being actively involved, the issue of information asymmetry is more severe in these inter-corporate loans, which in turn leads to higher costs of borrowing. For example, borrowers may be charged with higher interest rates or are more likely to be required to pledge collaterals to their loans. Therefore, in this paper, we focus on the mechanism of how firms mitigate asymmetric information in inter-corporate borrowings, in particular, role of cross-ownership between borrowers and lenders. Cross-ownership is an important form of economic organization that can help to mitigate information asymmetry, and in particular, weak legal protections such as property rights (Morck, Wolfenzon and Yeung, 2005).

Therefore, we hypothesize that cross-ownership can mitigate information asymmetry between borrowers and lenders, thus decreases the cost and collateral requirement of inter-corporate loans. This impact would be more pronounced if the borrowers and the lenders are cross-owned via majority stakes, where one party could directly influence the operation and decision making of the other party. We also posit that the beneficial role of cross-ownership on the inter-corporate loans is greater for the borrowing firms with poorer access to bank credit. These firms include non-state-owned firms, private firms and smaller firms. Lastly, we posit that the positive impact of cross-ownership is more pronounced for borrowers that operate in financially developed or legally strong provinces, as well as during times of loose monetary policy.

To empirically test these above hypotheses, we manually collect a unique set of 1,091 inter-corporate loans across 31 provinces in China between 2004 and 2014. Starting in 2004, listed firms started to disclose inter-corporate borrowing or lending activities in Shanghai and Shenzhen stock exchanges.

The empirical estimates show a strong effect of ownership structure on inter-corporate loans. Specifically, cross-owned firms borrow at significantly lower interest rates than non-related firms, and are also less likely to be required to pledge collaterals to their loans. Economically, the existence of cross-ownership linkage between borrowers and lenders reduces the average loan interest rates by 3.4 percent and lowers the collateral requirements by almost 50 percent. Our findings further show that the effect of cross-ownership on the cost and collateral requirement of inter-corporate loans is greater for firms with poorer credit access, i.e. non-state-owned firms, private firms and smaller firms. Cross-ownership is also more beneficial for firms operating in regions with less

developed financial markets or weaker legal protections, or during the periods of tight monetary policy. However, these heterogeneous impacts are only valid for the interest rates, not for the collateral requirements.

One possible concern in this paper is that inter-corporate loans with cross-ownership might not be exogenous. To formally address this issue, we apply an instrumental variable approach that exploits the non-lending related-party activities at the industry-year level as a plausibly exogenous instrument. Clearly, the non-lending related-party transactions can predict the occurrence of inter-corporate loans with cross-ownership linkages between borrowers and lenders. At the same time, these non-lending related-party activities are not directly correlated with the conditions of inter-corporate loans. Therefore, this instrument is valid and yields robust and consistent findings that cross-ownership reduces inter-corporate debt rates and collateral requirements by mitigating information asymmetry.

Our study thus contributes to the literature on informal financing. For example, Allen, Qian and Qian (2005) emphasize that, due to the imperfection of the legal system and that banks lack ability to screen and prevent risks, informal financing thus emerges in China based on relationships. Some other researches focus on the pricing of informal financing. Theoretically, Fabbri and Menichini (2010) present that both information superiority and the clearing advantage of the suppliers would lead to a decline in the pricing of trade credit, which is cheaper than bank credit. Empirically, using the US Small Business Survey, Giannetti, Burkart and Ellingsen (2011) find that most of the firms can get trade credit at very low prices. In the study of informal financing based on family-friend relationships, Lee and Person (2016) discover that most of the loans from family members or friends are extremely cheap, sometimes the real interest rates can even be negative. However, given the difficulties of obtaining informal financing data, the discussion on informal financing is not yet sufficient. Using a unique dataset of inter-corporate loans, this paper is able to provide direct evidence on the effect of cross-ownership linkages on the cost and the collateral requirement of inter-corporate loans.

This paper also adds to the literature on the internal capital market of business groups. Earlier work suggests that establishing business groups in the emerging markets is in response to the institutional drawbacks in these countries, i.e. information opacity and

weak legal protections (Shin and Stultz, 1998; Khanna and Yafeh, 2015). Establishing equity relationships help the member firms take funding from other members, which alleviates the issues of credit constraints and adverse external credit shocks (Gopalan, Nanda and Seru, 2007; Buchuk, Larrain, Muñoz and Urzúa, 2014). Due to the limited availability of reliable data, most of the previous literature does not directly examine the operation of the internal capital market within business groups. It is still vague on what determinants the loan conditions of intra-group lending. Our paper thus partially fills the gap that we provide direct comparison on the cost and collateral requirement between inter-corporate lending with or without cross-ownerships.

Finally, our results also contribute to the literature on the effect of ownership structure on firms' debt financing activities (La Porta, Lopez-de-Silanes and Zamarripa, 2003; Lin, Ma, Malatesta and Xuan, 2011). Our findings certify that ownership structure is indeed an important determinant of inter-corporate loan conditions, in addition to the related lending by banks. Our results indicate that cross-ownership decreases the cost and collateral requirement of inter-corporate loans and these effects are greater for the credit constrained firms. Therefore, cross-ownership mitigates information asymmetry and the potential conflicts in inter-corporate lending.

The rest of the paper is structured as follows. The financial market and background for inter-corporate loans are introduced in Section 2. Section 3 describes the data and reports the summary statistics. Section 4 develops the methodology and hypotheses. Section 5 reports the empirical results and Section 6 concludes.

2. Financial Market in China

The Chinese formal financial market is dominated by commercial banks (Allen, Qian and Qian, 2005). However, commercial banks in China favor lending to state-owned enterprises, publicly listed and larger firms, which discriminate against smaller and private firms (Firth, Lin and Wong, 2008). At the same time, the capital market, which includes the bond and equity markets, is developing rapidly. China's bond market has grown rapidly and is now the third largest in the world, behind only the US and Japan. By the end of 2015, the market value of the outstanding bonds in China accounts to almost 60 percent of the GDP, among which corporate bonds account to one thirds of total bond volumes. Similarly, Shanghai and Shenzhen stock exchanges grow rapidly

since their foundations in 1990. By the end of 2015, Shanghai and Shenzhen stock exchanges ranked 5th and 8th worldwide. However, major participants in the Chinese capital market are larger state-owned firms. Non-state-owned, private and relatively smaller firms are thus more credit constrained and often resort to alternative financing channels that are informal.

Shadow banking activities have been growing rapidly in China during recent years, in particular after 2010 when a series of restrictive policies were imposed by the China Banking Regulatory Commission (CBRC). These policies include the raising of the required reserve ratio, as well as the credit rationing and interest rate controls. In other words, different types of off-balance-sheet financing activities of banks are indirectly encouraged. Core Chinese shadow banking products were estimated at approximately US\$ 3.7 trillion at year-end 2012, accounting for 44 percent of China's GDP in the same year. According to S&P, shadow banking credit in China has been growing at an annual rate of 34% over recent years. Thus, although bank lending remains far bigger than the shadowy sort and is still expanding at an astonishing pace, its rate of growth has recently stabilised. The growth rate of shadow lending, in contrast, is accelerating. Shadow banks accounted for almost a third of the rise in lending in 2013, swelling by over 50% in the process.

A variety of shadow banking products can be found in China, ranging from the trust products, wealth management products to inter-corporate loans. Wealth management products have been the major source of funding, according to CBRC, with issuance estimated to be approximately US\$ 1.2 trillion at year-end 2012, a 55% increase over 2011. Wealth management products are often rolled over at short-term intervals, more and more new issuances are needed to pay off the expiring ones. As a result, a vicious cycle has developed, sparking growth in the shadow banking activities in China and rendering these types of investments highly vulnerable to a sudden shortage of funds due to the severe maturity mismatch.

Lots of attention about shadow banking in China also centered on trusts. By offering returns as high as 8 to 10 percent, they raise money from businesses and individuals frustrated by the low cap the government imposes for interest rates on bank deposits. The interest they charge to borrowers, naturally, is even higher. Trusts lend to firms that are unable to borrow from banks, often because they are in frothy industries, such

as property or steel, where regulators see signs of overinvestment and have instructed banks to curb lending.

The focus of this paper, inter-corporate loans, are yet another fast-growing form of the shadow banking activities. These loans often involve cash-rich companies, often well-connected state-owned enterprises, lending to less well-connected firms. Such loans, often made using banks as intermediaries to get around regulations forbidding such lending, expose the financial sector to yet more risk. The value of new inter-corporate loans in March 2014 was US\$ 39 billion, up US\$ 10 billion from the earlier year. In comparison, companies borrowed US\$ 117 billion via inter-corporate loans in the first three months of 2014; corporate bond issuance over the same period amounted to only US\$ 63 billion.

The shadow banking system in the US consists of securitized loans and obligations, as well as money market funds. In contrast, China's shadow banking system has direct links to commercial banks. Banks act as important distribution channels for shadowy financial products or as intermediaries for inter-corporate loans. Significant concerns have been raised about the interconnected relationship between banks and the shadow banking activities in China. A particular fear is that risk exposure might be heightened owing to a lack of transparency on these types of investments. The less regulated and more complicated shadow banking activities in China might create and give rise to system-wide financial instability.

Just as a crisis in shadow banking could spread to the real economy, a sharp downturn in some real sectors could also cause trouble for shadow banks, leading to a broader financial mess. Moreover, China still has imposed capital account control, i.e. control on both capital inflows and outflows. As foreign investors are more skeptical on the prospectus of the Chinese economy, there are concerns about the decreasing inflow of foreign capital. In addition, China is moving towards the opening of capital account and financial reform which might lead to the massive outflows of domestic capital to search for higher yields, thus the combined effects might dry up the local liquidity.

Overall, there is huge risk associated with China's shadow banking system, i.e. inter-corporate loans, which could lead to a financial distress or even crisis in China. China has pulled global growth in the last decades, attracting investments, exporting goods

all over the world, as well as financing businesses and governments. So, a slowdown in production and investment in China hampers global growth. Therefore, this paper focuses and thus sheds lights on a very important issue, not only for China, but also for other countries as well.

3. Data and Summary Statistics

Starting in 2004, publicly listed firms in China started to disclose their inter-corporate borrowing or lending activities in both Shanghai and Shenzhen stock exchanges. This enables us to manually collect detailed information on inter-corporate loans including loan conditions such as interest rate, maturity, and collateral, as well as information of borrowing and lending firms. Our final sample consists of 1,091 inter-corporate loans borrowed by 708 firms from 434 companies across 31 provinces in China from 2004 to 2014.² Figure 1 plots the geographical locations of the borrowers and lenders. Both the borrowers and the lenders spread across regions in China, but are more clustered in the more developed eastern coastal regions such Beijing, Shanghai and Guangdong.

[Inset Figure 1 here]

For public firms, we are able to obtain their information from their annual reports. For private firms, we collect their information either from their official websites or from information disclosed by their loan counter-parties. More importantly, we hand collect a firm's ownership information to identify the controlling and minority shareholders, as well as business group relationships. In addition, we also collect information on the institutional characteristics from the National Economic Research Institute (NERI), as well as the People's Bank of China (PBOC). Detailed variable definitions and sources are shown in Table 1.

[Inset Table 1 here]

3.1. Loan Cross-Ownership

To identify whether the borrower and lender of an inter-corporate loan are linked with

² The 31 provinces in China include: Anhui, Beijing, Chongqing, Fujian, Gansu, Guangdong, Guangxi, Guizhou, Hainan, Hebei, Heilongjiang, Henan, Hubei, Hunan, Jiangsu, Jiangxi, Jilin, Liaoning, Neimenggu, Ningxia, Qinghai, Shandong, Shanghai, Shannxi, Shanxi, Sichuan, Tianjin, Xinjiang, Xizang, Yunnan and Zhejiang.

cross-ownership, we construct a *Cross-ownership* dummy that equals 1 if cross owned and 0 otherwise. The cross-ownership of the borrowing and lending firms consists of two situations. First, the borrower and lender directly hold another party's equity and could thus exert influence on its business operations. Second, there is no direct equity holding between the borrower and the lender, but their equity shares are both held by a parent company (*Parent cross-ownership*). Among the 1,091 inter-corporate loans, 735 loans are between firms that are directly cross-owned, and only 62 of the loans occur between firms that are owned by a same parent enterprise. Among firms that are directly cross-owned, we further identify whether this cross-ownership is via majority or minority stake (shares above or below 50 percent). Of the 735 inter-corporate loans with cross-ownership, 287 loans occur via minority stake (*Minority cross-ownership*) while 448 through majority stake (*Majority cross-ownership*). Summary statistics are shown in Table 2.

[Inset Table 2 here]

Behind these averages lies substantial variation across regions in China. As shown in Table 3, borrowing from a cross-owned firm is more common in many provinces like Hebei, Heilongjiang, Jilin, Neimenggu, Shandong, Shannxi, Shanxi, Sichuan, Tianjin, Xinjiang and Xizang. In contrast, inter-corporate loans with cross-ownership happens less in provinces such as Anhui, Jiangsu and Zhejiang. There also exists substantial variation across regions in terms of majority or minority stakes.

[Inset Table 3 here]

3.2. Loan Characteristics

For each inter-corporate loan, we collect information on the loan conditions including the *Loan interest rate*, *Loan collateral*, *Loan maturity* as well as *Loan amount*. To be more specific, *Loan interest rate* is the all-in-one cost of a loan in percentage. *Loan collateral* is a dummy that equals 1 if the inter-corporate loan is collateralized and 0 otherwise. *Loan maturity* counts the maturity of the loan at initiation in years. Lastly, *Loan amount* presents the logarithm of the loan amount in RMB. As shown in Table 2, inter-corporate loans pay an average interest rate of almost 8 percent, which can be as high as 24 percent. In regions like Chongqing, Qinghai and Zhejiang, the average

interest rate amounts to more than 10 percent, whereas the interest rate is only about 5 percent in Heilongjiang, Jilin and Xinjiang. 37 percent of all inter-corporate loans are collateralized on average but also vary significantly across regions in China. A typical loan has an average maturity of 1.39 years but can be as long as 15 years. On average, an inter-corporate loan amounts 82 million RMB.

What's more, we also calculate the logarithm of the geographical distance between the borrower and the lender in inter-corporate loans (*Borrower-lender distance*). This is to control, for example, information asymmetry, spatial credit rationing, or lending relationships (i.e. Petersen and Rajan, 2002; Stein, 2002; Degryse and Ongena, 2005; Agarwal and Hauswald, 2010). We use the geo-coordinates of the borrowers and the lenders to calculate the distance. The average distance between the borrower and the lender in inter-corporate loans equals 70 kilometers.

3.3. Firm Characteristics

We obtain the borrowing firm's characteristics either through the inter-corporate loan announcements or via the annual reports of the publicly listed firms. Specifically, we distinguish whether a firm is owned by the Chinese government (*State owned firm*) or if the firm is listed in a stock exchange (*Publicly listed firm*). *Firm age* is defined as the number of years since a borrower has been registered in the State Administration for Industry and Commerce in China. Lastly, we construct *Firm asset* in logarithm. As many of the borrowers of inter-corporate loans are private companies, we are not able to collect complete information on their assets. For this reason, we do not incorporate *Firm asset* as a control variable in our main analysis, but only use it when examining the heterogeneity across firms. From our summary statistics as shown in Table 2, we find that almost 60 percent of the inter-corporate loans in our sample are borrowed by state owned firms (SOEs). This indicates that the government still plays an important role in the Chinese economy. However, among all the inter-corporate loans in China, only 17 percent of the loans are borrowed by public firms, whereas the rest 83 percent are all by the private firms. These private firms face tighter access to bank credit and therefore are more likely to seek to borrow from the inter-corporate loan market. The average firm has been established almost 10 years in the sample, with some firms that having been operated for more than five decades. Of all the 1,091 loans in our sample, we can get information on the borrower firm's asset for 833 of them. On average, firm

assets amount to 720 million RMB, which indicates that the firms in our sample are relatively larger firms.

3.4. Institutional Characteristics

We are interested in understanding the possibly heterogeneous impact of institutional characteristics such as financial development, legal environment, or monetary policies in the inter-corporate loan market. Therefore, we utilize the Index of Marketization of China's Provinces constructed by the National Economic Research Institute (NERI) to measure both the financial development and the legal environment across provinces in China. Specifically, we construct a dummy variable, *Financially developed province*, that equals 1 if the index of a province's financial development belongs to the top ten in China and 0 otherwise. Similarly, *Legally strong province* is a dummy that equals 1 if the legal environment index of a province ranks top ten in China and 0 otherwise. In addition, we also measure the tightness of the monetary policy in China in each year. *Loose monetary policy* is a dummy that equals 1 when the monetary policy in China is loose (with a monetary policy index above 50) and 0 otherwise. This monetary policy index is obtained from the People's Bank of China (PBOC), which is the central bank in China. On average, 73 percent of the inter-corporate loans occur in provinces with higher level of financial development. About half of these loans are in provinces with better legal environment. 80 percent of borrowings happen when the monetary policy is loose, that is, when borrowing is less costly.

4. Methodology and Hypothesis

To investigate the role of cross-ownership in reducing information asymmetry in the inter-corporate loans, we focus on two aspects. We start with exploring whether firms are able to borrow at lower (favorable) interest rates from other cross-owned firms in the inter-corporate loan market. Then we further check if an inter-corporate borrower is less likely to be required to pledge a collateral when borrowing from a lender with cross-ownership. Inter-corporate loans occur when a company provides the funds but, to circumvent a ban on direct lending to other firms, it designates a commercial bank to lend the money to a specific borrower. The lender firm stipulates the amount, tenor, and rate of the loan, while the banks only earn fees from both sides without the loans showing up on their balance sheets. Therefore, information asymmetry is more severe

in the inter-corporate loan market than in the bank lending market. In particular, we conjecture that corporations rely on cross-ownerships between borrowers and lenders to mitigate information asymmetry, which is a vital form of economic organization that helps mitigating weak legal protections (Morck, Wolfenzon and Yeung, 2005). So our first hypothesis is:

H1: Cross-ownership can mitigate the information asymmetry between borrowers and lenders, thus decreases the cost of inter-corporate loans and collateral requirement.

In order to test this first hypothesis, we seek to model (1) to as shown below:

$$\text{Loan conditions}_{isrt} = \alpha_s + \alpha_r + \alpha_t + \beta \text{Cross_ownership}_{isrt} + \delta X_{isrt} + \varepsilon_{isrt} \quad (1)$$

where $\text{Loan conditions}_{isrt}$ represents either *Loan interest rate* or *Loan collateral* of an inter-corporate loan i in sector s in region r in year t . X_{isrt} represents the control variables, including other loan characteristics (e.g. *Loan amount*, *Loan maturity* and *Borrow-lender distance*) augmented by the firm characteristics (e.g. *State owned firm*, *Publicly listed firm* and *Firm age*), institutional structures (e.g. *Financially developed province*, *Legally strong province* and *Loose monetary policy*). We also apply a set of industry, region and year fixed effects given by α_s , α_r and α_t , respectively. Therefore, instead of adding controls for the time-invariant sector and province level features, we employ both sector and region fixed effects to control for all these variables that may affect the inter-corporate loan interest rates or the collateral requirements. Finally, we employ year fixed effects to further control for all the common time-varying factors. $\text{Cross_ownership}_{isrt}$ is our variable of interest which captures cross-ownership links between the borrowers and the lenders in the inter-corporate loans.

Secondly, the impact of cross-ownership on the cost or collateral requirement of inter-corporate loans would be more pronounced if borrowers and lenders are connected by stronger ownership linkages, i.e. via majority stakes that one party is able to directly insert influence the operation and decision making of the other party. Thus our second hypothesis is as follows:

H2: The effect of cross-ownership on inter-corporate loans is more pronounced for if the borrowers and lenders are connected via stronger ownership linkages.

To examine the second hypothesis, we adjust model (1) to include cross-owned firms via majority stakes, as shown below:

$$\text{Loan conditions}_{isrt} = \alpha_s + \alpha_r + \alpha_t + \beta_1 \text{Cross_ownership}_{isrt} + \beta_2 \text{Majority cross_ownership}_{isrt} + \delta X_{isrt} + \varepsilon_{isrt} \quad (2)$$

where $\text{Cross_ownership}_{isrt}$ refers to a dummy that equals 1 if the borrowers and the lenders are cross-owned and $\text{Majority cross_ownership}_{isrt}$ is a dummy that equals 1 if the borrowers and the lenders are cross-owned via majority stakes. If the second hypothesis holds, we would expect a significant estimate of β_2 that has the same sign as β_1 . We apply the same set of control variables, as well as industry, region and year fixed effects as in model (1).

The third hypothesis posits that the effect of cross-ownership on the inter-corporate loans is more pronounced for borrowing firms with poorer access to bank credit. For example, Bai, Lu and Tao (2006) show that both small and medium size enterprises (SMEs) and non-state-owned enterprises (non-SOEs) face discriminations when they borrow from the state-owned banks in China. In addition, Petersen and Rajan (2002) present that the non-SOEs, private firms, and SMEs face higher informational costs in accessing to bank credit. This leads to our second hypothesis:

H3: The effect of cross-ownership on inter-corporate loans is more pronounced for firms with poorer access to formal credit, i.e. non-state-owned firms, private firms, or smaller firms.

To test this hypothesis, we estimate the following regression:

$$\text{Loan conditions}_{isrt} = \alpha_s + \alpha_r + \alpha_t + \beta_1 \text{Cross_ownership}_{isrt} + \beta_2 \text{Cross_ownership}_{isrt} * \text{Firm characteristics}_{isrt} + \delta X_{isrt} + \varepsilon_{isrt} \quad (3)$$

where $\text{Firm characteristics}_{isrt}$ refers firm characteristics including whether a firm is a *State owned firm*, a *Publicly listed firm*, as well as the age of the firm (*Firm age*). The same set of control variables, as well as industry, region and year fixed effects are

included in the regression.

Lastly, we posit that the positive impact of cross-ownership on inter-corporate loans is more pronounced when borrower firms face less favorable institutional environment. To measure the institutional characteristics, we focus on three main aspects, namely, the development of local financial market, the soundness of local legal environment and the tightness of monetary policy. These institutional factors could affect a firm's access to credit. For example, Levine (1997) presents that, due to reduced information asymmetry, firms in more developed financial markets have better access to credit. Regarding legal environment, La Porta, Lopez-de-Silanes, Shleifer and Vishny (1997) show that better creditor protections are associated with broader credit markets. Later, Levine (1998, 1999), Djankov, McLiesh and Shleifer (2007), Bae and Goyal (2009) and Weill (2011) confirm that stronger creditor protection leads to more bank credit while poor law enforcement hampers bank lending. Therefore, our last hypothesis is:

H4: The effect of cross-ownership on inter-corporate loans is more pronounced for firms with poorer access to formal credit, i.e. firms that located in the financially less developed provinces or where legal protection is weaker, as well as during the periods of tight monetary policy.

To test the final hypothesis, we estimate the model as follows:

$$\text{Loan conditions}_{isrt} = \alpha_s + \alpha_r + \alpha_t + \beta_1 \text{Cross_ownership}_{isrt} + \beta_2 \text{Cross_ownership}_{isrt} * \text{Institutional characteristics}_{rt} + \delta X_{isrt} + \varepsilon_{isrt} \quad (4)$$

where *Institutional characteristics*_{rt} presents regional characteristics that whether a province is a *Financially developed province*, or a *Legally strong province*, as well as an indicator whether a specific time period experiences *Loss monetary policy*. Once again, we include a same set of control variables, as well as industry, region and year fixed effects.

5. Empirical Results

5.1. Baseline Estimates

This section starts with the baseline estimates. But before moving to the multivariate

regressions, we first point to Table 4 that provides a vivid illustration that link cross-ownership between the borrowers and the lenders with inter-corporate loan conditions including the interest rates and collateral requirements. Borrower-lender relationships are categorized in two main groups: the firms with cross-ownerships (797 firms) and those without cross-ownerships (294 firms). Among firms with cross-ownerships, we further classify firms into three categories: cross-ownerships via minority stakes (287 firms), via majority stakes (448 firms) or via the same parent or holding company (62 firms). We find that borrowers with cross-ownerships pay significantly lower interest rates (6.52 percent) than firm without such connections (11.71 percent). Similarly, those cross-owned borrowers are about 70 percent less likely to be required to pledge collaterals to their inter-corporate loans. All of these differences are highly significant under a two-sample t-test with unequal variances. So, at a first sight, cross-ownerships mitigate information asymmetry between the borrowers and lenders, therefore lower the interest rate and collateral requirement.

[Inset Table 4 here]

We then formally test the relationship between cross-ownership and loan conditions in the inter-corporate loan market. The results are presented in Table 5. The dependent variable is the *Loan interest rate* in columns (1) to (5) while the *Loan collateral* is the dependent variable in columns (6) to (10). We start with the most basic specification that only include *Cross-ownership* as the explanatory variable and gradually add a set of control variables at the firm level in columns (2) and (7), at the institutional level in columns (3) and (8), and at the loan level in columns (4) and (9). Lastly, sector, region and year fixed effects are further included in specifications (5) and (10). Across all the specifications, we discover that cross-ownerships between the borrowers and lenders significantly lowers the interest rates and collateral requirements in the inter-corporate loans. Economically, based on the most sophisticated specifications in models (5) and (10), borrowers with cross-ownerships with the lenders pay 3.4 percent lower interest rates and are 50 percent less likely to be required to pledge collaterals.

The estimation coefficients of the control variables are consistent with the literature. For example, state-owned firms and publicly listed firms pay lower interest rates and are also less likely to pledge collaterals. Interest rates are generally lower in times of loss monetary policy and in financially more developed regions, borrowers have more

collaterals. Furthermore, the interest rates are higher for larger loans and loans with shorter maturity, as well as for loans with collaterals.

[Inset Table 5 here]

However, the findings in Table 5 may be biased due to the endogeneity of the cross-ownerships between borrowers and lenders. It might be the case that the firms without cross-ownership affiliations in the inter-corporate loans are different from firms with cross-ownerships. We, therefore, first address the endogeneity issue by stricter control of the fixed effects. The results are reported in columns (1), (2), (5) and (6) in Table 6. In columns (1) and (5), instead of industry, province and year fixed effects, we control for the industry-year, province-year and industry-province interacted fixed effects. By doing this, we can control all the time varying industry and province level drivers, as well as time-invariant industry-province level variables. Furthermore, in columns (2) and (6), we include the borrower fixed effects to rule out any time-invariant borrower specific unobservables. The results stay significantly negative both for interest rates and collateral requirements, indicating that our findings are robust to stricter control of fixed effects. In addition, we cluster the standard errors by borrowers in columns (3) and (7) and by provinces in columns (4) and (8). This is to allow for the possibility that regression errors are correlated within the same province or the same borrower. Our main findings remain the same.

[Inset Table 6 here]

To further deal with the endogeneity issue, we also seek to the instrumental variable approach. Specifically, we exploit non-lending related-party activities at the industry-year level as a plausibly exogenous instrument for cross-ownerships in inter-corporate loans. This is because, clearly, non-lending related-party transactions at the industry-year level are able to predict the occurrence of inter-corporate loans among borrowers and lenders with cross-ownerships. Intuitively, if the firm is operating in an industry where related-party activities are common in a specific year, this firm is more likely to borrow from related lenders. However, at the same time, these non-lending related-party activities should not be directly correlated with the interest rates or the collateral requirements of the inter-corporate loans. Results are presented in Table 7 where first-stage results are shown in columns (1)-(2) and corresponding second-stage estimates

are in columns (3)-(4) respectively. From the first-stage results we can find that firms are more likely to borrow from cross-owned lenders if their industry of operation has more non-lending related-party activities. Second-stage estimates confirm the findings that cross ownership reduces inter-corporate debt cost and collateral requirement.

[Inset Table 7 here]

Therefore, the results in Tables 4-7 support our first hypothesis that cross-ownerships can mitigate information asymmetry between borrowers and lenders, thus reduces the cost of inter-corporate loans and collateral requirements.

5.2. Degree of Cross-Ownership

This section focuses on our second hypothesis that the impact of cross-ownerships is more pronounced for borrowers and lenders with stronger ownership linkages, that is, cross-owned via majority stakes. To formally test this hypothesis, we point to Table 8 where both *Cross-ownership* and *Majority cross-ownership* dummies are included in the regressions.

Columns (1)-(5) show the results for *Loan interest rate* and columns (6)-(10) present the findings for *Loan collateral*. We find that cross-ownership significantly lowers the interest rates (by 3.2 percent) and the collateral requirements (by 48 percent) in inter-corporate loans, which is consistent with our baseline findings. Then, conditional on being cross-owned, those borrowers that are cross-owned via majority stakes with the lenders pay even lower interest rates and are even less likely to pledge collaterals. The results are also economically relevant. For example, those cross-owned borrowers via majority stakes pay almost 0.5 percent lower interest rates than cross-ownerships via minority stakes or via a same parent company. At the same time, these firms are also 4 percent less likely to be required to pledge collaterals to their loans (although results are not always statistically significant).

[Inset Table 8 here]

In a word, the results in this section confirm our second hypothesis that the effect of cross-ownership on inter-corporate loans is more pronounced for if the borrowers and lenders are connected via stronger ownership linkages.

5.3. Heterogeneity across Firm Characteristics

The effect of cross-ownership on inter-corporate loans might differ across firms with different levels of credit access. Intuitively, as cross-ownership significantly mitigates the issue of information asymmetry and hence lowers the interest rates and collateral requirements, its impact should be more pronounced for firms with poorer access to credit. As shown by literature, non-state-owned firms, private firms, and smaller firms face higher informational costs in accessing to credit. To formally test this hypothesis, we seek to model (3) and the results are reported in Table 9. In general, we find the same results that cross-ownerships significantly lowers the interest rates and collateral requirements in the inter-corporate loans. State-owned firms and publicly listed firms are also found to pay lower interest rates and are less likely to be requested to pledge collaterals to their loans. Moreover, larger firms also face favorable interest rates with similar collateral requirements. The main results are reported by the interaction terms. Interestingly, we find that there is heterogeneous effect of cross-ownership on the cost of inter-corporate loans, but not on the collateral requirements. In detail, the beneficial effect of cross-ownership is less pronounced for the state-owned firms, publicly listed firms and larger firms. Economically, state-owned firms pay, on average, 2.4 percent higher interest rates, and public firms pay 2 percent extra. In addition, a one standard deviation increase in asset (1.86) is related with 0.5 percent increase in loan rates.

[Inset Table 9 here]

These findings partially confirm our hypothesis that the impact of cross-ownership on inter-corporate loans is more pronounced for firms with poorer credit access, which is only valid for the interest rates.

5.4. Heterogeneity across Institutional Characteristics

Finally, we investigate the heterogeneous effect of cross-ownership on inter-corporate loans in relation to institutional characteristics. Specifically, we focus on the financial market development and legal environment on the provincial level, as well as macro monetary policy tightness on the annual level. These measures capture the tightness of credit access in general. For example, due to reduced information asymmetry, firms in more developed financial markets have better access to credit. Likewise, the stronger

legal protection leads to more bank credit while poor law enforcement hampers bank lending. Lastly, when monetary policy is loose, firms would gain better credit access.

Table 10 presents the estimation results. We find again that the heterogeneity across institutional environment is only valid for the interest rates, but not for the collateral requirements. Specifically, firms in financially developed or legally strong provinces pay relatively higher interest rates. Similarly, in times of loose monetary policy, firms also face higher loan costs. Thus these results indicate that the beneficial role of cross-ownership is more pronounced when institutional environment is worse and confirms our fourth hypothesis.

[Inset Table 10 here]

6. Conclusions

In this paper we examine the inter-corporate loans, which is a main form of informal debt financing in China. This issue is important as the rapid growth and development of the inter-corporate loan market has raised concerns on the shadow banking system, and consequently the financial stability in China, which is vital globally in a financial inter-dependent world.

We focus on the prevalence of cross-ownership affiliation between the borrowers and the lenders of inter-corporate loans on loan conditions including the interest rates and collateral requirements. This study, therefore, helps to improve the understanding of the mechanism of the informal financing in the emerging markets.

To formally test this issue, we manually collect a unique set of 1,091 inter-corporate loans across 31 provinces in China between 2004 and 2014. We find that the presence of cross-ownership significantly lowers the interest rates and collateral requirements of inter-corporate loans. The results are also economically relevant. Cross-ownership affiliation significantly lowers interest rates by 3.4 percent and decreases the collateral requirements by almost 50 percent. Furthermore, we find that the beneficial effect of cross-ownership affiliation on inter-corporate loans is greater for those borrowers with poorer access to credit, i.e. non-state-owned firms, private firms, and smaller firms, as well as firms operating in provinces with less developed financial markets and weaker

legal protections, or in times of tight monetary policy.

Our research is also very policy relevant. We provide direct evidence that ownership structure plays an important role in informal financing. The results also shed lights on the demand side of credit in the informal financing market as well as the intermediary role of corporations in supplying credit through inter-corporate loans. These findings have important implications for the financial stability and economic development, not only for China, for also for other countries in a financial inter-dependent world.

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Table 1. Variable Definitions and Sources

This table includes the variable definitions and sources. NERI is the National Economic Research Institute in China. PBOC is the People's Bank of China (Central Bank in China).

Variable	Definitions	Sources
<i>Loan Cross-Ownership</i>		
Cross-ownership	dummy =1 if borrowers and lenders are cross owned directly or by a parent company, 0 otherwise	Authors' collection
Minority cross-ownership	dummy =1 if borrowers and lenders are cross owned directly via minority stake, 0 otherwise	Authors' collection
Majority cross-ownership	dummy =1 if borrowers and lenders are cross owned directly via majority stake, 0 otherwise	Authors' collection
Parent cross-ownership	dummy =1 if borrowers and lenders are cross owned by a parent company, 0 otherwise	Authors' collection
<i>Loan Characteristics</i>		
Loan interest rate	interest rate of a inter-corporate loan in percentage	Authors' collection
Loan collateral	dummy =1 if a inter-corporate loan is collateralized, 0 otherwise	Authors' collection
Loan maturity	maturity of a inter-corporate loan at initiation in years	Authors' collection
Loan amount	log of a inter-corporate loan amount in RMB	Authors' collection
Borrower-lender distance	log of the distance between a borrower and a lender in kilometers	Authors' collection
<i>Firm Characteristics</i>		
State owned firm	dummy =1 if a borrower is a state-owned firm, 0 otherwise	Authors' collection
Publicly listed firm	dummy =1 if a borrower is a publicly listed firm, 0 otherwise	Authors' collection
Firm age	age of a borrower firm in years	Authors' collection
Firm asset	log of a borrower firm's asset in RMB	Authors' collection
<i>Institutional Characteristics</i>		
Financially developed province	dummy = 1 if the province of a borrower ranked top 10 in the financial development index in China, 0 otherwise	NERI
Legally strong province	dummy = 1 if the province of a borrower ranked top 10 in the legal environment index in China, 0 otherwise	NERI
Loose monetary policy	dummy = 1 when monetary policy is loose (monetary policy index above 50), 0 otherwise	PBOC
<i>Instrumental Variable</i>		
Non-lending related-party activity	industry-year average of the ratio of non-lending related-party transactions over total revenue	Authors' collection

Table 2. Summary Statistics

This table reports the summary statistics for all the variables. Definitions and sources of the variables are provided in Table 1.

Variable	Obs.	Mean	Std.	Min.	Max.
<i>Loan Cross-Ownership</i>					
Cross-ownership	1,091	0.73	0.44	0	1
Minority cross-ownership	1,091	0.26	0.44	0	1
Majority cross-ownership	1,091	0.41	0.49	0	1
Parent cross-ownership	1,091	0.06	0.23	0	1
<i>Loan Characteristics</i>					
Loan interest rate	1,091	7.92	3.82	0.00	24.00
Loan collateral	1,091	0.37	0.48	0	1
Loan maturity	1,091	1.39	1.19	0.08	15.00
Loan amount	1,091	18.22	1.28	13.12	21.82
Borrower-lender distance	1,091	4.26	2.38	0.00	8.11
<i>Firm Characteristics</i>					
State owned firm	1,091	0.59	0.49	0	1
Publicly listed firm	1,091	0.17	0.38	0	1
Firm age	1,091	9.70	7.06	0	60
Firm asset	833	20.39	1.86	14.72	26.98
<i>Institutional Characteristics</i>					
Financially developed province	1,091	0.73	0.44	0	1
Legally strong province	1,091	0.54	0.50	0	1
Loose monetary policy	1,091	0.80	0.40	0	1
<i>Instrumental Variable</i>					
Non-lending related-party activity	1,091	0.03	0.08	-0.36	0.85

Table 3. Cross-Ownership and Inter-corporate Loans: Across Countries

This table shows the average percentages of cross-ownership, as well as loan interest rate and collateral across provinces in China. Table 1 contains all definitions and Table 2 the summary statistics for each included variable.

Province	Cross-ownership				Loan		Number of Loans
	Overall	Minority	Majority	Parent	Interest rate	Collateral	
Anhui	50.00%	16.67%	33.33%	0.00%	9.35%	60.00%	30
Beijing	77.59%	44.83%	22.41%	10.34%	7.52%	24.14%	58
Chongqing	58.33%	12.50%	25.00%	20.83%	10.23%	87.50%	24
Fujian	71.70%	52.83%	9.43%	9.43%	6.91%	35.85%	53
Gansu	50.00%	0.00%	50.00%	0.00%	8.41%	100.00%	2
Guangdong	74.03%	25.97%	40.26%	7.79%	7.72%	33.77%	77
Guangxi	77.78%	33.33%	22.22%	22.22%	7.84%	16.67%	18
Guizhou	80.00%	12.00%	52.00%	16.00%	7.72%	20.00%	25
Hainan	71.43%	71.43%	0.00%	0.00%	9.49%	71.43%	7
Hebei	94.50%	10.09%	81.65%	2.75%	6.18%	3.67%	109
Heilongjiang	100.00%	0.00%	100.00%	0.00%	5.65%	18.18%	11
Henan	80.00%	40.00%	40.00%	0.00%	6.96%	40.00%	50
Hubei	67.74%	16.13%	45.16%	6.45%	8.80%	45.16%	31
Hunan	56.52%	26.09%	21.74%	8.70%	8.28%	78.26%	23
Jiangsu	46.15%	20.51%	24.36%	1.28%	9.18%	58.97%	78
Jiangxi	84.62%	38.46%	46.15%	0.00%	7.07%	38.46%	13
Jilin	100.00%	22.22%	77.78%	0.00%	5.68%	0.00%	9
Liaoning	88.24%	17.65%	64.71%	5.88%	6.86%	17.65%	17
Neimenggu	93.33%	33.33%	46.67%	13.33%	6.35%	6.67%	15
Ningxia	75.00%	50.00%	25.00%	0.00%	6.92%	25.00%	8
Qinghai	66.67%	16.67%	50.00%	0.00%	10.83%	33.33%	6
Shandong	94.74%	26.32%	61.40%	7.02%	6.38%	35.09%	57
Shanghai	64.81%	38.89%	18.52%	7.41%	8.20%	38.89%	54
Shannxi	92.86%	35.71%	50.00%	7.14%	7.42%	35.71%	14
Shanxi	95.00%	32.50%	62.50%	0.00%	6.32%	25.00%	40
Sichuan	94.74%	40.35%	49.12%	5.26%	7.26%	19.30%	57
Tianjin	100.00%	0.00%	80.00%	20.00%	9.22%	6.67%	15
Xinjiang	100.00%	55.00%	25.00%	20.00%	5.58%	20.00%	20
Xizang	100.00%	0.00%	100.00%	0.00%	6.37%	0.00%	3
Yunnan	82.35%	29.41%	41.18%	11.76%	8.14%	17.65%	17
Zhejiang	39.33%	13.33%	26.00%	0.00%	10.54%	69.33%	150

Table 4. Cross-Ownership and Inter-corporate Loans: Univariate Results

This table reports univariate results on the relationship between cross-ownership and inter-corporate loan conditions. *, **, *** indicate significance at the 10%, 5% and 1% level, respectively, for a two-sample t-test of a difference in means with unequal variances. For the t-tests we compare loan conditions among firms with and without cross-ownership. Table 1 contains all definitions and Table 2 the summary statistics for each included variable.

	<i>Loan</i>		<i>Number of Loans</i>
	Interest rate	Collateral	
<i>Cross-ownership</i>	6.52%***	18.44%***	797
Minority cross-ownership	6.83%***	18.12%***	287
Majority cross-ownership	6.23%***	16.74%***	448
Parent cross-ownership	7.16%***	32.26%***	62
<i>No cross-ownership</i>	11.73%	89.12%	294
<i>Total</i>	7.92%	37.49%	1,091

Table 5. Cross-Ownership and Inter-corporate Loans: Baseline Estimates

This table shows the regressions to estimate the relationship between cross-ownership and loan conditions for inter-corporate loans. Table 1 contains all definitions and Table 2 the summary statistics for each included variable. Coefficients are listed in the first row, p-values are reported below in the brackets, and the corresponding significance levels are placed adjacently. *** Significant at 1%, ** significant at 5%, * significant at 10%.

Dependent Variable	Loan interest rate					Loan collateral				
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
Cross-ownership	-5.206*** [0.000]	-4.406*** [0.000]	-4.421*** [0.000]	-3.876*** [0.000]	-3.395*** [0.000]	-0.707*** [0.000]	-0.603*** [0.000]	-0.590*** [0.000]	-0.540*** [0.000]	-0.497*** [0.000]
State owned firm		-1.409*** [0.000]	-1.244*** [0.000]	-1.170*** [0.000]	-1.008*** [0.000]		-0.164*** [0.000]	-0.149*** [0.000]	-0.131*** [0.000]	-0.107*** [0.000]
Publicly listed firm		-0.487* [0.067]	-0.611** [0.023]	-0.666** [0.015]	-0.245 [0.394]		-0.115*** [0.000]	-0.132*** [0.000]	-0.123*** [0.000]	-0.149*** [0.000]
Firm age		-0.013 [0.370]	-0.010 [0.470]	-0.011 [0.413]	-0.015 [0.245]		-0.001 [0.714]	-0.001 [0.603]	-0.001 [0.662]	-0.001 [0.542]
Financially developed province			0.445* [0.086]	0.430* [0.098]	0.743 [0.366]			0.110*** [0.000]	0.106*** [0.001]	0.121 [0.243]
Legally strong province			-0.082 [0.732]	-0.111 [0.642]	1.390 [0.409]			-0.006 [0.847]	-0.005 [0.868]	-0.023 [0.914]
Loose monetary policy			-0.996*** [0.000]	-0.987*** [0.000]	-0.448 [0.443]			-0.023 [0.404]	-0.011 [0.691]	-0.078 [0.293]
Loan amount				0.162** [0.030]	-0.020 [0.785]				-0.002 [0.797]	-0.006 [0.498]
Loan maturity				-0.176** [0.026]	-0.136* [0.077]				-0.008 [0.413]	-0.014 [0.143]
Loan collateral				0.748*** [0.003]	0.494** [0.044]					
Loan interest rate									0.011*** [0.003]	0.008** [0.044]
Borrower-lender distance				-0.040 [0.297]	-0.005 [0.902]				0.001 [0.775]	-0.005 [0.334]
Year Fixed Effects	No	No	No	No	Yes	No	No	No	No	Yes
Province Fixed Effects	No	No	No	No	Yes	No	No	No	No	Yes
Industry Fixed Effects	No	No	No	No	Yes	No	No	No	No	Yes
R-squared	0.366	0.396	0.407	0.415	0.507	0.419	0.450	0.458	0.461	0.511
Observations	1,091	1,091	1,091	1,091	1,091	1,091	1,091	1,091	1,091	1,091

Table 6. Cross-Ownership and Inter-corporate Loans: Robustness Checks

This table shows the robustness checks of the relationship between cross-ownership and loan conditions for inter-corporate loans. Table 1 contains all definitions and Table 2 the summary statistics for each included variable. Coefficients are listed in the first row, p-values are reported below in the brackets, and the corresponding significance levels are placed adjacently. *** Significant at 1%, ** significant at 5%, * significant at 10%.

Dependent Variable Model	Loan interest rate				Loan collateral			
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Cross-ownership	-3.154*** [0.000]	-1.706** [0.010]	-3.395*** [0.000]	-3.395*** [0.000]	-0.492*** [0.000]	-0.497*** [0.000]	-0.497*** [0.000]	-0.497*** [0.000]
Control Variables	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Year Fixed Effects	No	Yes	Yes	Yes	No	Yes	Yes	Yes
Province Fixed Effects	No	No	Yes	Yes	No	No	Yes	Yes
Industry Fixed Effects	No	No	Yes	Yes	No	No	Yes	Yes
Province*Year Fixed Effects	Yes	No	No	No	Yes	No	No	No
Province*Industry Fixed Effects	Yes	No	No	No	Yes	No	No	No
Industry*Year Fixed Effects	Yes	No	No	No	Yes	No	No	No
Firm Fixed Effects	No	Yes	No	No	No	Yes	No	No
Firm Clustering	No	No	Yes	No	No	No	Yes	No
Province Clustering	No	No	No	Yes	No	No	No	Yes
R-squared	0.601	0.915	0.507	0.507	0.553	0.829	0.511	0.511
Observations	1,091	1,091	1,091	1,091	1,091	1,091	1,091	1,091

Table 7. Cross-Ownership and Inter-corporate Loans: Instrumental Variable

This table shows the instrumental variable regressions to estimate the relationship between cross-ownership and loan conditions for inter-corporate loans. Marginal effects are reported for the first-stage results. Table 1 contains all definitions and Table 2 the summary statistics for each included variable. Coefficients are listed in the first row, p-values are reported below in the brackets, and the corresponding significance levels are placed adjacently. *** Significant at 1%, ** significant at 5%, * significant at 10%.

Dependent Variable	First-stage		Second-stage	
	Cross-ownership		Loan interest	Loan collateral
Model	(1)	(2)	(3)	(4)
Non-lending related-party activity	0.478*** [0.001]	0.412*** [0.010]		
Cross-ownership			-4.712*** [0.000]	-0.410*** [0.000]
Control Variables	Yes	Yes	Yes	Yes
Year Fixed Effects	Yes	Yes	Yes	Yes
Province Fixed Effects	Yes	Yes	Yes	Yes
Industry Fixed Effects	Yes	Yes	Yes	Yes
Observations	1,091	1,091	1,091	1,091

Table 8. Cross-Ownership and Inter-corporate Loans: Degree of Cross-Ownership

This table shows the regressions to estimate the relationship between cross-ownership, majority cross-ownership and loan conditions for inter-corporate loans. Table 1 contains all definitions and Table 2 the summary statistics for each included variable. Coefficients are listed in the first row, p-values are reported below in the brackets, and the corresponding significance levels are placed adjacently. ***

Significant at 1%, ** significant at 5%, * significant at 10%.

Dependent Variable	Loan interest rate					Loan collateral				
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
Cross-ownership	-4.839*** [0.000]	-3.920*** [0.000]	-3.967*** [0.000]	-3.480*** [0.000]	-3.179*** [0.000]	-0.685*** [0.000]	-0.562*** [0.000]	-0.553*** [0.000]	-0.511*** [0.000]	-0.481*** [0.000]
Majority cross-ownership	-0.652*** [0.003]	-0.832*** [0.000]	-0.787*** [0.000]	-0.744*** [0.001]	-0.476** [0.033]	-0.039 [0.140]	-0.071*** [0.009]	-0.063** [0.021]	-0.056** [0.041]	-0.036 [0.198]
State owned firm		-1.329*** [0.000]	-1.175*** [0.000]	-1.109*** [0.000]	-0.975*** [0.000]		-0.157*** [0.000]	-0.144*** [0.000]	-0.127*** [0.000]	-0.105*** [0.000]
Publicly listed firm		-0.798*** [0.004]	-0.898*** [0.001]	-0.939*** [0.001]	-0.399 [0.177]		-0.141*** [0.000]	-0.155*** [0.000]	-0.144*** [0.000]	-0.161*** [0.000]
Firm age		-0.015 [0.270]	-0.013 [0.361]	-0.014 [0.310]	-0.017 [0.198]		-0.001 [0.609]	-0.001 [0.519]	-0.001 [0.576]	-0.001 [0.492]
Financially developed province			0.387 [0.134]	0.383 [0.140]	0.597 [0.468]			0.105*** [0.001]	0.102*** [0.001]	0.110 [0.291]
Legally strong province			-0.070 [0.766]	-0.104 [0.662]	1.286 [0.445]			-0.005 [0.871]	-0.004 [0.881]	-0.030 [0.887]
Loose monetary policy			-0.983*** [0.000]	-0.972*** [0.000]	-0.450 [0.441]			-0.022 [0.425]	-0.011 [0.700]	-0.078 [0.292]
Loan amount				0.152** [0.042]	-0.019 [0.795]				-0.003 [0.744]	-0.006 [0.503]
Loan maturity				-0.181** [0.021]	-0.140* [0.069]				-0.008 [0.383]	-0.015 [0.135]
Loan collateral				0.689*** [0.006]	0.471* [0.055]					
Loan interest rate									0.010*** [0.006]	0.008* [0.055]
Borrower-lender distance				-0.043 [0.264]	-0.004 [0.908]				0.001 [0.814]	-0.005 [0.337]
Year Fixed Effects	No	No	No	No	Yes	No	No	No	No	Yes
Province Fixed Effects	No	No	No	No	Yes	No	No	No	No	Yes
Industry Fixed Effects	No	No	No	No	Yes	No	No	No	No	Yes
R-squared	0.371	0.403	0.413	0.420	0.508	0.420	0.453	0.460	0.462	0.511
Observations	1,091	1,091	1,091	1,091	1,091	1,091	1,091	1,091	1,091	1,091

Table 9. Cross-Ownership and Inter-corporate Loans: Heterogeneity across Firms Characteristics

This table shows the regressions to estimate the heterogeneous relationship between cross-ownership and loan conditions for inter-corporate loans across firm characteristics. Table 1 contains all definitions and Table 2 the summary statistics for each included variable. Coefficients are listed in the first row, p-values are reported below in the brackets, and the corresponding significance levels are placed adjacently. *** Significant at 1%, ** significant at 5%, * significant at 10%.

Dependent Variable	Loan interest rate			Loan collateral		
Model	(1)	(2)	(3)	(5)	(6)	(7)
Cross-ownership	-4.035*** [0.000]	-3.548*** [0.000]	-8.532*** [0.000]	-0.471*** [0.000]	-0.510*** [0.000]	0.038 [0.912]
Cross-ownership * State owned firm	2.428*** [0.000]			-0.083 [0.157]		
Cross-ownership * Publicly listed firm		2.015** [0.022]			0.161 [0.150]	
Cross-ownership * Firm asset			0.291*** [0.005]			-0.027 [0.111]
State owned firm	-2.810*** [0.000]	-1.023*** [0.000]	-0.826*** [0.000]	-0.044 [0.395]	-0.108*** [0.000]	-0.092*** [0.002]
Publicly listed firm	-0.385 [0.176]	-2.095** [0.015]	-0.417 [0.129]	-0.144*** [0.000]	-0.296*** [0.006]	-0.144*** [0.001]
Firm asset			-0.340*** [0.000]			0.008 [0.565]
Control Variables	Yes	Yes	Yes	Yes	Yes	Yes
Year Fixed Effects	Yes	Yes	Yes	Yes	Yes	Yes
Province Fixed Effects	Yes	Yes	Yes	Yes	Yes	Yes
Industry Fixed Effects	Yes	Yes	Yes	Yes	Yes	Yes
R-squared	0.520	0.509	0.608	0.511	0.511	0.515
Observations	1,091	1,091	833	1,091	1,091	833

Table 10. Cross-Ownership and Inter-corporate Loans: Heterogeneity across Institutional Characteristics

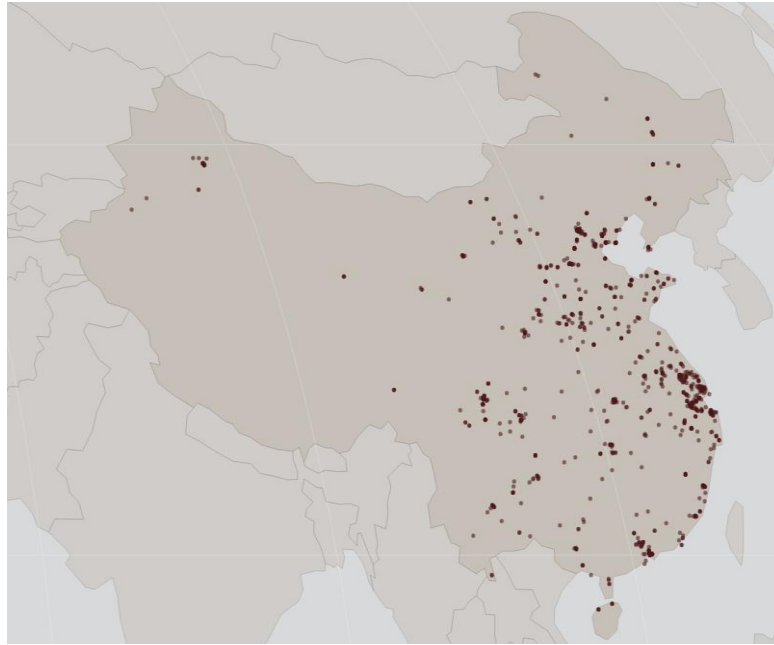
This table shows the regressions to estimate the heterogeneous relationship between cross-ownership and loan conditions for inter-corporate loans across institutional characteristics. Table 1 contains all definitions and Table 2 the summary statistics for each included variable. Coefficients are listed in the first row, p-values are reported below in the brackets, and the corresponding significance levels are placed adjacently. *** Significant at 1%, ** significant at 5%, * significant at 10%.

Dependent Variable Model	Loan interest rate			Loan collateral		
	(1)	(2)	(3)	(5)	(6)	(7)
Cross-ownership	-4.455*** [0.000]	-4.223*** [0.000]	-4.608*** [0.000]	-0.425*** [0.000]	-0.471*** [0.000]	-0.487*** [0.000]
Cross-ownership * Financially developed province	1.271** [0.021]			-0.085 [0.225]		
Cross-ownership * Legally strong province		1.168** [0.010]			-0.037 [0.527]	
Cross-ownership * Loose monetary policy			1.522*** [0.002]			-0.013 [0.832]
Financially developed province	-0.204 [0.824]	0.744 [0.364]	0.503 [0.540]	0.184 [0.113]	0.121 [0.244]	0.123 [0.238]
Legally strong province	1.341 [0.425]	0.511 [0.765]	1.441 [0.390]	-0.020 [0.925]	0.004 [0.984]	-0.023 [0.912]
Loose monetary policy	-0.533 [0.362]	-0.541 [0.355]	-1.381** [0.035]	-0.072 [0.333]	-0.075 [0.314]	-0.070 [0.403]
Control Variables	Yes	Yes	Yes	Yes	Yes	Yes
Year Fixed Effects	Yes	Yes	Yes	Yes	Yes	Yes
Province Fixed Effects	Yes	Yes	Yes	Yes	Yes	Yes
Industry Fixed Effects	Yes	Yes	Yes	Yes	Yes	Yes
R-squared	0.509	0.509	0.511	0.511	0.510	0.510
Observations	1,091	1,091	1,091	1,091	1,091	1,091

Figure 1. Plots of Borrowers and Lenders of Inter-Corporate Loans in China

This map plots the borrowers and lenders of inter-corporate loans in China. Panel A plots the borrower firms and Panel B plots the lender firms.

Panel A. Borrower Firms



Panel B. Lender Firms

